

University of Maryland case studies: your review tasks

NCLEX NGN Case Study Studio · September 30, 2026 · 20 case studies (CS2–CS21) · 162 review points

These case studies were converted from the medical-surgical faculty case studies of the Maryland Next Gen NCLEX Test Bank Project (University of Maryland School of Nursing, nursing.umaryland.edu/mnwc/initiatives/nextgen-nclex/nextgen-nclex-library/). Asthma is CS1 (already in NURS 1021 Unit 3). Every screen carries an acknowledgment of the source and author. All of them are in the question bank under **Others, hidden from students** until you publish them.

What I checked for each item: every screen renders in the player and scores full marks with its answer key; the readiness check finds nothing missing (except the draft flag); the chart never loses a tab or an entry from one screen to the next; opening and saving in the editor changes nothing.

How to review and publish a case

1. Open the studio (**?author=1**), sign in, and go to **Question bank**. Set the status filter to **Hidden from students** or search for “University of Maryland”.
2. Open the case and go through the items for that case in this document (tick each box). Use **Preview with answers** to read each screen as a student sees it.
3. Make any changes in the editor and save (the chart carries changes forward to later screens).
4. If you want the case in a course unit instead of **Others**, change the Course and Unit chips.
5. Publish: click the **Ready for students** chip → **Show to students** (or ⋯ → Show to students in the question bank). Do the same for its stand-alone trend or bow-tie.

Changes made to every case

- Laboratory values are in SI units (Canada). The author’s reference ranges were converted, not replaced, so each question keys exactly as written. Blood gases stay in mm Hg (as MCC prints them). Temperatures are shown in °C with °F.
- Answer options were reordered where a correct answer was listed first. Where a source’s answer key was missing or unreadable, the key was taken from its rationale; this is flagged in that case.
- Clear errors (doses, units, lab values that contradict the question) were corrected and are listed below. Content that is debatable was kept as the author wrote it and flagged for your decision.
- Each source’s stand-alone trend or bow-tie was added as its own stand-alone question, unless it repeated screen 6.

Overview

Case	Topic	Items	Review points	Status
CS2	Tuberculosis	Case study + Trend	4	In the bank, hidden (draft)
CS3	Acute Respiratory Failure	Case study	8	In the bank, hidden (draft)

CS4	Breast Cancer	Case study + Trend	10	In the bank, hidden (draft)
CS5	Burns	Case study	8	In the bank, hidden (draft)
CS6	Chest Pain (MI)	Case study + Bowtie	8	In the bank, hidden (draft)
CS7	Compartment Syndrome	Case study + Trend	9	In the bank, hidden (draft)
CS8	COPD I	Case study	8	In the bank, hidden (draft)
CS9	COPD II	Case study + Bowtie	7	In the bank, hidden (draft)
CS10	Deep Vein Thrombosis II	Case study + Bowtie	7	In the bank, hidden (draft)
CS11	End Stage Renal Disease and Dialysis	Case study + Bowtie	8	In the bank, hidden (draft)
CS12	Gastroesophageal Reflux Disease	Case study + Bowtie	7	In the bank, hidden (draft)
CS13	Heart Failure	Case study + Bowtie	9	In the bank, hidden (draft)
CS14	HIV with an Opportunistic Infection	Case study + Trend	8	In the bank, hidden (draft)
CS15	Type II Diabetes and Ketoacidosis	Case study + Bowtie	10	In the bank, hidden (draft)
CS16	Liver Failure	Case study + Bowtie	7	In the bank, hidden (draft)
CS17	Prostate Cancer with Prostatectomy	Case study + Bowtie	8	In the bank, hidden (draft)
CS18	Care of Client Post-Op Spine Surgery (Laminectomy)	Case study + Bowtie	8	In the bank, hidden (draft)
CS19	Stroke	Case study + Bowtie	10	In the bank, hidden (draft)
CS20	Tension Pneumothorax	Case study + Bowtie	10	In the bank, hidden (draft)
CS21	Thyroid Storm	Case study + Bowtie	8	In the bank, hidden (draft)

University of Maryland - CS2: Tuberculosis

Source: "Tuberculosis.docx" (medical-surgical) · Author: Elizabeth Mackessy-Lloyd, DNP, RN, CNE, Hood University · In the bank, hidden (draft)

What to review

- ☐ Isoniazid order: the source lists 1000 mg PO daily, above the 300 mg maximum daily dose; shown as 300 mg PO daily (the dose used in the 4-month rifapentine-moxifloxacin regimen). Please confirm the regimen and dose.
- ☐ Laboratory values shown in SI units with the author's reference ranges converted (not replaced), so every question keys as written.
- ☐ CS2 Trend: the source's trend is a grid (improved / declined / unchanged), so it is a matrix question. Its rationale said the pulse oximetry reading "has increased", but it is 95% on room air at both visits and is keyed Unchanged; the rationale now says unchanged.
- ☐ Quoted client statements were fixed in the database (they showed “ codes); check they read correctly.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS2 case_1790769567553

Screen 1 · Select N (3)

Select the 3 assessment findings that are most concerning at this time.

- ✓ Lung sounds
- ✓ Temperature
- ✓ Pain in chest

Screen 2 · Matrix (one per row)

For each assessment finding, click to specify if the finding is a risk factor or is not a risk factor for tuberculosis infection.

- Client is overweight → Not a Risk Factor
- Medication list → Not a Risk Factor
- Recent travel to Asia → Risk Factor
- Extended travel with a group → Risk Factor
- Temperature → Risk Factor
- Blood pressure → Not a Risk Factor
- Cough → Risk Factor
- Pain characteristics → Risk Factor

Screen 3 · Drag and drop

Drag the most appropriate word choice from the list of options to complete the following sentence.

- The nurse should recognize that the client's priority problem to address at this time is [Risk for spreading infection].

Screen 4 · Matrix (one per row)

For each potential nursing intervention, click to specify whether the intervention is appropriate or not appropriate to include in the plan of care.

- Implement a fluid restriction while taking tuberculosis medication → Not Appropriate
- Initiate a 24-hour urine collection → Not Appropriate
- Instruct the client to wear a mask when around others for the next 3 weeks → Appropriate
- Teach coughing and deep breathing techniques → Appropriate
- Alert the client's family that she has tuberculosis → Not Appropriate
- Encourage a high-carbohydrate diet → Not Appropriate
- Withhold opioid pain medication → Not Appropriate
- Apply an N95 face mask → Appropriate

Screen 5 · Matrix (one per row)

For each client teaching point, click to specify whether the information needs to be provided immediately, at the time of client follow-up, or is not indicated for this client.

- “Your cholesterol is high, and we need to discuss your diet.” → On Return Visit
- “You will need to take the medications every day for at least 4 months.” → Immediately
- “If you continue to cough at home, please save all of your sputum.” → Not Indicated
- “You will need to have regular labs to check the effects of this medication on your liver.” → Immediately
- “Try to pace your activities to preserve your energy.” → Immediately
- “Tuberculosis is not spread by germs on dishes or linens.” → Immediately
- “Your A1C is high, and you may have diabetes.” → On Return Visit

Screen 6 · Dyad

Complete the following sentence by choosing from the lists of options.

- The nurse determines the client's status is [improving]. The nurse should now [review medication teaching].

University of Maryland - CS2 Trend standalone_1790776800001

Question · Matrix (one per row)

For each finding, click to specify if the finding indicates the client's status has improved, declined, or is unchanged.

- Missing several doses of medication → Declined
- Pulse oximetry reading → Unchanged
- Productive cough → Unchanged
- Blood pressure → Declined
- Temperature → Improved
- Fatigue → Unchanged
- WBC count → Improved
- Pain characteristics → Declined

University of Maryland - CS3: Acute Respiratory Failure

Source: "Acute-Respiratory-Distress.docx" (medical-surgical) · Author: Kadriyya Clark, DNP, RN, CNE, Community College of Baltimore County · In the bank, hidden (draft)

What to review

- ☐ Amoxicillin IVPB is prescribed: there is no IV amoxicillin in Canada/US, and it is a narrow choice for gram-negative bacteremia. Consider another antibiotic (e.g. piperacillin-tazobactam or ceftriaxone).
- ☐ Blood culture reads "gram-negative cocci" (unusual; gram-negative rods are more typical). Please confirm.
- ☐ 0900 blood gas: SaO₂ 96% with PaO₂ 75 mm Hg is inconsistent. Please review.
- ☐ Lactate 4.5 mmol/L is an urgent (sepsis) finding but is not keyed on screen 1. Consider adding it to the key or changing the value.
- ☐ PEEP 5 "mmHg" changed to 5 cm H₂O. Screen 1 option "PCO₂ 50 mm Hg" changed to 51 mm Hg (the value in the chart).
- ☐ Screen 5 answer key taken from the source's highlighted Key table (IV 0.9% sodium chloride, midazolam, chest X-ray).
- ☐ The source's stand-alone trend is the same question as screen 6, so it was not added separately.
- ☐ Laboratory values in SI units (magnesium 1.5 mEq/L → 0.75 mmol/L; cell counts × 10⁹/L); blood gases stay in mm Hg.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS3 case_1790776800002

Screen 1 · Select N (4)

Select the 4 findings that are most urgent.

- ✓ pH 7.20
- ✓ PaCO₂ 51 mm Hg
- ✓ Respiratory rate 30
- ✓ Pulse oximetry reading 87%

Screen 2 · Matrix (multiple per row)

For each finding, click to specify if the finding is consistent with acute respiratory failure or pneumonia. Each finding may support more than one condition.

- Crackles in bilateral lower lobes → Acute Respiratory Failure, Pneumonia
- Pulse oximetry reading 87% on 100% oxygen → Acute Respiratory Failure, Pneumonia
- PaCO₂ 51 mm Hg → Acute Respiratory Failure
- pH 7.20 → Acute Respiratory Failure

Screen 3 · Multiple choice

What does the nurse determine is the priority for this client's care?

- ✓ Intubating for mechanical ventilation support

Screen 4 · Matrix (one per row)

For each potential intervention, click to specify whether the intervention is indicated or not indicated to include in the plan of care.

- Administer sedatives → Indicated
- Repeat chest X-ray → Indicated
- Schedule suctioning every 2 hours → Not Indicated
- Administer amiodarone → Not Indicated
- Administer IV antibiotics → Indicated
- Position supine with head midline → Not Indicated
- Obtain an electrocardiogram (ECG) → Indicated

Screen 5 · Highlight

Click to highlight the 3 orders the nurse should implement first.

- ✓ Start IV of 0.9% sodium chloride at 75 mL/hr
- ✓ Midazolam 2–4 mg IV push every 1 hour as needed for agitation
- ✓ Call for chest X-ray

Screen 6 · Matrix (one per row)

For each finding, click to specify if the finding indicates that the client's status has improved, declined, or is unchanged.

- Heart rate → Improved
- Temperature → Unchanged
- Respiratory rate → Unchanged
- Blood pressure → Improved
- Pulse oximetry reading → Declined
- Agitation → Improved

University of Maryland - CS4: Breast Cancer

Source: "Breast-Cancer.docx" (medical-surgical) · Author: Kimberly Allen, DNP, RN, CNE; Stephanie Howard, MS, CRNP, FNP-C; Lisa Seldomridge, PhD, RN, CNE; Salisbury University · In the bank, hidden (draft)

What to review

- ☐ Screen 2 (matrix, more than one answer per row): the author's key marks numbness/tingling for lymphedema AND DVT, and HR 104 for cellulitis only, but the author's rationale said numbness is "only associated with lymphedema" and that tachycardia "could be associated with DVT". The key was kept; the rationale was rewritten to match it. Please confirm the key.
- ☐ Screen 2: "Sodium 148 mmol/L" keyed as consistent with lymphedema (rationale: hypernatremia could add to the swelling) is a weak link. Consider replacing that row.
- ☐ Screen 4: amoxicillin is keyed as an anticipated order "to treat a possible infection that triggered the lymphedema", although the client is afebrile with no redness (cellulitis not suspected). The client is also neutropenic (WBC $1.5 \times 10^9/L$). Please confirm the antibiotic order and rationale.
- ☐ Discharge prescriptions (1400 note): acetaminophen 500 mg PO twice daily is an unusual frequency for pain (usually every 4-6 hours as needed). Please review.
- ☐ Screen 6: the source's rationale repeated screen 5; it was rewritten for the statements actually asked (ice bag, antibiotics for a cut, aching and heaviness, resuming exercises). "Ice bag = no understanding" follows the usual advice to avoid temperature extremes on the affected arm; please confirm.
- ☐ Screen 1: WBC $1.5 \times 10^9/L$ is keyed as one of the 4 most significant findings (not explained in the source rationale; a sentence was added).
- ☐ Lab values converted to SI with the author's ranges: glucose 68 mg/dL → 3.8 mmol/L (< 5.5); Hgb 11 g/dL → 110 g/L; Hct 33% → 0.33 L/L; albumin 2.8 g/dL → 28 g/L (35-55); WBC/platelets $\times 10^9/L$; K+/Na+ mmol/L.
- ☐ Screen 3: "drag 1 condition and 1 cause" is a drag-and-drop with both blanks scored together (1 point, "rationale rule" in the source).
- ☐ Screen 4 options were reordered so a correct answer is not listed first.
- ☐ Trend (stand-alone): uses the source's own flow sheet, where the 1200 BP is 126/72 (screen 6 of the case shows 132/72, as in the source). No question depends on it.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS4 case_1790776800004

Screen 1 · Select N (4)

Select the 4 assessment findings that are most significant.

- ✓ R arm numbness and tingling
- ✓ R arm pain rated 5/10
- ✓ WBC $1.5 \times 10^9/L$
- ✓ R arm swelling

Screen 2 · Matrix (multiple per row)

For each finding, click to indicate if the finding is consistent with cellulitis, lymphedema, or deep vein thrombosis (DVT). Each finding may support more than one condition.

- Right arm swelling → Cellulitis, Lymphedema, DVT
- Right arm numbness/tingling → Lymphedema, DVT
- Temperature 98°F (36.6°C) → Lymphedema, DVT
- HR 104 → Cellulitis
- Sodium 148 mmol/L → Lymphedema
- Albumin 28 g/L → Lymphedema

Screen 3 · Drag and drop

Drag 1 condition and 1 cause to fill in the blanks of the following sentence.

- The greatest risk for this client is developing [Chronic lymphedema] due to [Post-op R radical mastectomy and axillary node dissection].
- (Scored as a pair: 1 point only if both blanks are correct.)

Screen 4 · Select all that apply

What orders does the nurse anticipate including in the plan of care? Select all that apply.

- ✓ Acetaminophen PO
- ✓ Diet as tolerated
- ✓ Sequential pneumatic compression
- ✓ Amoxicillin PO
- ✓ Physical therapy

Screen 5 · Matrix (one per row)

For each potential nursing action, click to specify whether the action is appropriate or not appropriate to include in the plan of care.

- Measure circumference of R wrist, forearm, and upper arm → Appropriate
- If needed, collect fingerstick blood glucose from R index finger → Not Appropriate
- Take blood pressure in L arm → Appropriate
- Teach client manual lymphatic drainage techniques → Not Appropriate
- Elevate R arm on two pillows → Appropriate
- Apply warm compresses to R arm → Not Appropriate
- Encourage oral fluids → Appropriate

Screen 6 · Matrix (one per row)

For each client statement, click to specify whether the statement indicates an understanding or no understanding of the teaching provided.

- "If my arm hurts, I'll use an ice bag on it." → No Understanding
- "It's good to use a gentle moisturizer on my arms every day." → Understanding
- "I will wear gloves when I work in the garden." → Understanding
- "If I get a cut on my right hand, I will start taking antibiotics immediately." → No Understanding
- "Losing weight might help the swelling." → Understanding
- "Some aching and heaviness in my right arm is expected." → No Understanding
- "I can resume all upper body exercises as soon as I want." → No Understanding
- "I will try to wear bracelets on my left arm only." → Understanding

Question · Matrix (one per row)

For each finding, click to specify if the finding indicates that the client's status has improved, declined, or is unchanged since admission to the clinic.

- Temperature → Unchanged
- Pain → Improved
- HR → Improved
- RR → Improved
- R upper arm circumference → Unchanged
- Anxiety → Declined

University of Maryland - CS5: Burns

Source: "Burn.docx" (medical-surgical) · Author: Stacy McGrath, EdD, MSN, RN, Salisbury University · In the bank, hidden (draft)

What to review

- ☐ Total body surface area: the source says 36% TBSA for burns to the front of both lower extremities; by the rule of nines, the anterior surfaces of both legs are about 18%. The 14,400 mL Parkland calculation ($4 \text{ mL} \times 100 \text{ kg} \times 36\%$) depends on 36%. Consider changing the burn description (e.g. front and back of both legs) so the numbers agree.
- ☐ Intake and Output: at 2200 the urine output was 10 mL/hr, which under the titration order ($< 15 \text{ mL/hr}$: increase 20%) should raise the rate at 2300 ($1320 \rightarrow \text{about } 1584 \text{ mL/hr}$), but the chart keeps 1320 mL/hr. Kept as in the source; please review.
- ☐ Fluid order wording: the source says "give half 8 hours since burn event and the second half over remaining 24 hours"; it now reads "over the first 8 hours ... over the remaining 16 hours" (the Parkland formula).
- ☐ Vital signs: screens 3 and 4 of the source label the second column 2030 (2100 elsewhere); 2100 is used throughout.
- ☐ Screen 1 ("drag the 4 findings") is a Select 4 question in the app (same content and 0/1 scoring).
- ☐ Screen 5 rationale: a sentence tying "increase the IV rate" to the urine-output titration order was added.
- ☐ Albumin order: "systolic B/P<80, diastolic B/P<50" now reads "systolic BP < 80 or diastolic BP < 50"; please confirm "or" vs "and".
- ☐ No laboratory values in this case (no SI conversion needed). The document's trend is the same question as screen 6, so no stand-alone was added.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS5 case_1790776800005

Screen 1 · Select N (4)

Select the 4 findings that are most urgent to follow up.

- ✓ Respirations
- ✓ Blood pressure
- ✓ Urine output
- ✓ Heart rate

Screen 2 · Matrix (multiple per row)

For each client finding, click to specify whether the finding is consistent with compartment syndrome, inhalation injury, or distributive shock. Each finding may support more than one condition.

- Urine output → Distributive Shock
- Tachypnea → Inhalation Injury, Distributive Shock
- Blood pressure → Distributive Shock
- Peripheral edema → Compartment Syndrome, Distributive Shock
- Pain → Compartment Syndrome

Screen 3 · Dyad

Complete the following sentences by choosing from the lists of options.

- The client is at most risk for complications associated with [distributive shock]. The client care priority is to [increase fluids].

Screen 4 · Matrix (one per row)

What orders does the nurse anticipate including in the plan of care? For each potential order, click to specify whether the order is indicated or not indicated.

- Obtain a blood gas → Indicated
- Double the IV infusion rate → Not Indicated
- Monitor basic metabolic panel → Indicated
- Obtain a chest X-ray → Indicated
- Culture wounds → Not Indicated

Screen 5 · Multiple choice

What action should the nurse take first?

- ✓ Increase the IV rate

Screen 6 · Matrix (one per row)

For each finding, click to indicate if the finding indicates the client's condition has improved, declined, or is unchanged.

- Blood pressure → Unchanged
- Heart rate → Unchanged
- Urine output → Improved
- Oxygen saturation → Declined
- Breath sounds → Declined
- Temperature → Unchanged

University of Maryland - CS6: Chest Pain (MI)

Source: "Chest-Pain-(MI).docx" (medical-surgical) · Author: Kadriyya Clark, DNP, RN, CNE, Community College of Baltimore County · In the bank, hidden (draft)

What to review

- ☐ Screen 3 has NO answer key in the source. It was keyed as ST-elevation myocardial infarction + coronary artery "blockage" (the rationale ends with the clot causing total blockage of the artery), but the rationale also describes plaque "rupture". Please confirm blockage vs rupture.
- ☐ Oxygen: the case applies oxygen at SpO2 93% on room air, and the bow-tie keys "Administer oxygen" as a correct action (source rationale: SpO2 should be $\geq 95\%$). Current ACS guidelines (AHA/CCS) recommend oxygen only when SpO2 is $< 90\%$. Consider revising the bow-tie key or the SpO2 value; the sentence about $\geq 95\%$ was removed from the rationale.
- ☐ Bow-tie: "Administer low-dose aspirin" is keyed incorrect (source: aspirin should be given in higher doses). Chewable aspirin 160-325 mg is standard; the option wording may confuse students. Please review.
- ☐ Orders: enoxaparin 100 mg SC (client 127 kg; with fibrinolysis the usual regimen is 30 mg IV bolus + 1 mg/kg SC, max 100 mg for the first doses), tenecteplase 50 mg IV (weight ≥ 90 kg, correct), metoprolol 25 mg PO, morphine 2 mg IV every 15 min PRN. Please review the enoxaparin order.
- ☐ Screen 5: "within 30 minutes" is the door-to-needle target for fibrinolysis; for an anterior STEMI, primary PCI within 90 minutes (door-to-balloon) is preferred when available. The question works as written; consider adding that the facility has no PCI capability.
- ☐ Screen 2: the row "ECG changes" now reads "ST elevation in leads V3 and V4" (NSTEMI and unstable angina can also show ECG changes, e.g. ST depression). CK-MB and troponin rows now say "Elevated".
- ☐ Labs in SI with the author's ranges: cholesterol 250 mg/dL $\rightarrow$ 6.5 mmol/L; magnesium 1.5 mEq/L $\rightarrow$ 0.75 mmol/L; troponin T 0.5 ng/mL $\rightarrow$ 500 ng/L (0-40); WBC/platelets $\times 10^9$ /L; K+/Na+ mmol/L. CK-MB kept in U/L as in the source.
- ☐ Screen 1 options were reordered so a correct answer is not listed first.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS6 case_1790776800006

Screen 1 · Select N (2)

Select the 2 findings that are most significant.

- ✓ ECG
- ✓ Troponin T

Screen 2 · Matrix (multiple per row)

For each client finding, click to specify whether the finding is consistent with ST-elevation myocardial infarction (STEMI), non-ST-elevation myocardial infarction (NSTEMI), or unstable angina. Each finding may be consistent with more than one condition.

- Elevated creatine kinase MB $\rightarrow$ STEMI, NSTEMI
- Crushing chest pain $\rightarrow$ STEMI, NSTEMI, Unstable Angina
- ST elevation in leads V3 and V4 $\rightarrow$ STEMI
- Elevated cholesterol $\rightarrow$ STEMI, NSTEMI, Unstable Angina
- Obesity $\rightarrow$ STEMI, NSTEMI, Unstable Angina
- Elevated troponin T $\rightarrow$ STEMI, NSTEMI

Screen 3 · Dyad

Complete the following sentence by choosing from the lists of options.

- The nurse should recognize that the client is most likely experiencing a/an [ST-elevation myocardial infarction] caused by coronary artery [blockage].

Screen 4 · Matrix (one per row)

For each potential medication order, click to specify whether the medication is indicated or not indicated to include in the plan of care.

- Analgesics → Indicated
- Antibiotics → Not Indicated
- Anticoagulants → Indicated
- Beta blockers → Indicated
- Fluid bolus → Not Indicated
- Fibrinolytics → Indicated

Screen 5 · Dyad

Complete the following sentence by choosing from the lists of options.

- The nurse should next give the [tenecteplase 50 mg IV], which should be given [within 30 minutes of admission].

Screen 6 · Matrix (one per row)

For each finding, click to specify if the finding indicates that the client's status has improved, declined, or is unchanged.

- Pulse → Declined
- Blood pressure → Improved
- Pulse oximetry reading → Improved
- Respiratory rate → Unchanged
- Pain → Improved

University of Maryland - CS6 Bowtie standalone_1790776801061

Question · Bow-tie

Complete the diagram by dragging from the choices below to specify what condition the client is most likely experiencing, 2 actions the nurse should take to address that condition, and 2 parameters the nurse should monitor to assess the client's progress.

- Condition: Myocardial infarction
- Actions: Administer sublingual nitroglycerin; Administer oxygen
- Parameters: Pain; Pulse oximetry reading

University of Maryland - CS7: Compartment Syndrome

Source: "Compartment-Syndrome.docx" (medical-surgical) · Author: Dawn Leukhardt, MSN, RN, College of Southern Maryland, La Plata, Maryland · In the bank, hidden (draft)

What to review

- ☐ Screen 4: the source is a grouped multiple-response question (Nursing / Medication / Collaborative, "each category may have more than one order"). The app has no grouped type, so it is a Select All with the category at the start of each option (same key, +/- scoring).
- ☐ Screen 5 (delegation): "Obtain vital signs every hour for 4 hours" is keyed as delegable, but the 1130 STAT order is vital signs every 15 minutes, and the client is hypotensive (BP 82/44). Consider rewording the option to match the order, and whether VS for an unstable client should be delegated.
- ☐ Chart timing: the source's screen 5 flow sheet already shows the 1145 and 1245 readings (after cast removal); in the app they appear on screen 6, with the 1145/1245 notes.
- ☐ Flow sheet 1245 color: the case shows N (normal) but the trend shows P (pale) for the same time. Screen 6 keys color as Improved (based on N). Please confirm.
- ☐ Screen 6 key: Pain/Motion/Color improved while the limb is now cold and pulseless; the trend then asks students to call this "deteriorating". Consider whether screen 6 should say so in its rationale.
- ☐ 0830 note: RR 29 (tachypnea on admission) is not addressed anywhere; possibly a typo for 20.
- ☐ Screen 3 rationale referred to abnormal vital signs and tachycardia that are not yet charted at 1115 (only in the 1115 note shown from screen 4); it was rewritten around the neurovascular findings.
- ☐ Screen 1 options were reordered so a correct answer is not listed first.
- ☐ No laboratory values in this case.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS7 case_1790776800007

Screen 1 · Select all that apply

Which findings require immediate follow-up? Select all that apply.

- ✓ Toes on the affected extremity are cool to the touch
- ✓ Right toes have noticeable pallor compared with the left
- ✓ Client describes absent sensation of the right lower leg
- ✓ Capillary refill greater than 3 seconds in the right toes
- ✓ Client reports pain of 10/10 in the R lower extremity

Screen 2 · Matrix (one per row)

For each potential complication of a fracture, click to indicate if the complication is an acute complication or a chronic complication associated with fractures.

- Venous thromboembolism → Acute Complication
- Complex regional pain syndrome → Chronic Complication
- Avascular necrosis → Chronic Complication
- Osteomyelitis → Chronic Complication
- Delayed union → Chronic Complication
- Compartment syndrome → Acute Complication

Screen 3 · Dyad

Complete the following sentence by choosing from the lists of options.

- Based on the assessment findings, the client is most likely experiencing [compartment syndrome], as most evidenced by the [neurovascular assessment].

Screen 4 · Select all that apply

Based on the assessment findings and recent vital signs, which orders does the nurse anticipate being included in the plan of care? Select all that apply.

- ✓ Nursing: Change diet to nothing by mouth
- ✓ Nursing: Cast removal saw to bedside STAT
- ✓ Nursing: Order compartment pressure measurement device to bedside
- ✓ Medication: Administer morphine 4 mg IV push for pain
- ✓ Medication: Administer IV normal saline bolus of 500 mL over 30 minutes
- ✓ Collaborative: Consult surgery team for possible fasciotomy

Screen 5 · Select all that apply

Which tasks can the nurse delegate to the unlicensed assistive personnel (UAP)? Select all that apply.

- ✓ Obtain vital signs every hour for 4 hours
- ✓ Assist client to use bedpan to void
- ✓ Assist client to maintain right leg at the level of the heart
- ✓ Measure intake and output
- ✓ Document client's height and weight

Screen 6 · Matrix (one per row)

For each finding, click to specify if the finding indicates that the client's status has improved, declined, or is unchanged since the onset of symptoms.

- Pain → Improved
- Motion → Improved
- Sensation → Unchanged
- Capillary refill → Unchanged
- Color → Improved
- Warmth → Declined
- Pulse → Declined

University of Maryland - CS7 Trend standalone_1790776801071

Question · Dyad

Complete the following sentences by choosing from the lists of options.

- Based on the data, the nurse determines the client's status is [deteriorating]. The nurse should now [notify the physician].

University of Maryland - CS8: COPD I

Source: "COPD.docx" (medical-surgical) · Author: Suzana Jarquin, MSN, RN, CNEcl, Frederick Community College · In the bank, hidden (draft)

What to review

- ☐ Oxygen: the client's SpO2 falls from 91% to 88% on ROOM AIR and no oxygen is ever started. Controlled oxygen (target 88-92%) is standard in a COPD exacerbation; consider adding low-flow oxygen to the 1015 note or orders, or explaining why it is withheld.
- ☐ Screen 2 keys digital clubbing as consistent with COPD. Clubbing is not a typical feature of COPD itself (it suggests another cause, such as lung cancer or bronchiectasis). Please review this row.
- ☐ Screen 4 keys "Encourage use of incentive spirometer" as correct during an acute exacerbation with severe dyspnea; please confirm. (Source is a grouped Nursing/Medication question; it is a Select All with the category at the start of each option.)
- ☐ Medication orders: acetaminophen 500 mg PO once "PRN for moderate to severe back pain" (the client has no back pain, and acetaminophen is usually for mild pain/fever); please review.
- ☐ Vital signs: the source labels the third column 1015 again; it is 1045 here (matching the 1045 note).
- ☐ Screen 5 rationale contained leftover text ("Albuterol was just givenfor Q4 you can make ,"); cleaned up. The antibiotic sentence now says it can be given after the corticosteroid.
- ☐ Screen 1 options were reordered so a correct answer is not listed first.
- ☐ No laboratory values in this case. The document's trend repeats screen 6, so no stand-alone was added.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS8 case_1790776800008

Screen 1 · Select all that apply

Which priority findings require immediate follow-up? Select all that apply.

- ✓ Accessory muscle use
- ✓ Coarse crackles
- ✓ Green-tinged sputum
- ✓ Tachypnea

Screen 2 · Matrix (multiple per row)

For each client finding, click to indicate if the finding is consistent with asthma, chronic obstructive pulmonary disease (COPD), pneumonia, or pulmonary embolism. Each finding may support more than one condition.

- Dyspnea → Asthma, COPD, Pneumonia, Pulmonary Embolism
- Productive cough → Asthma, COPD, Pneumonia
- Barrel chest appearance → COPD
- Digital clubbing → COPD
- Expiratory wheezing → Asthma, COPD, Pneumonia

Screen 3 · Multiple choice

Which condition is the client most likely experiencing?

- ✓ COPD exacerbation

Screen 4 · Select all that apply

Which orders should the nurse include in the plan of care at this time? Select all that apply.

- ✓ Nursing: Place client in a high-Fowler's position
- ✓ Nursing: Encourage client use of incentive spirometer
- ✓ Medication: Ipratropium bromide and albuterol
- ✓ Medication: Methylprednisolone

Screen 5 · Multiple choice

Which medication should the nurse give next?

- ✓ Methylprednisolone sodium succinate

Screen 6 · Matrix (one per row)

For each client finding, click to specify if the finding indicates that the client's status has improved, declined, or is unchanged.

- Respiratory rate → Declined
- Heart rate → Declined
- Pulse oximetry reading → Declined
- Blood pressure → Declined
- Wheezing → Improved
- Breath sounds → Declined
- Temperature → Unchanged

University of Maryland - CS9: COPD II

Source: "COPD-II.docx" (medical-surgical) · Author: Mary DiBartolo, PhD, RN-BC, CNE, FGSA, FAAN, Salisbury University · In the bank, hidden (draft)

What to review

- ☐ Discharge orders: levofloxacin 500 mg PO daily (community-acquired pneumonia is usually treated with 750 mg daily, or 500 mg in older regimens) and roflumilast 500 mcg PO daily (often started at 250 mcg for 4 weeks). Please review the antibiotic dose and duration (none given).
- ☐ Screen 2: the source row "Pulse oximeter 87%" was changed to 85% (the admission value in the chart; 87% appears only at 1400 on screen 5). Its rationale also listed "dyslipidemia" as a COPD risk factor, which is not a row and is not a recognized risk factor; that phrase was removed.
- ☐ Screen 2 keys clubbing as associated with COPD (as in CS8, clubbing is not typical of COPD alone). Please review.
- ☐ Home inhaler: the source says salmeterol/fluticasone "250/25"; the product is labelled fluticasone/salmeterol 250/50 (Diskus) or salmeterol/fluticasone 25/250 (MDI). Written here as 25/250; please confirm.
- ☐ Client Information shows the name "Roy Blue" as in the source (fictional).
- ☐ Labs in SI with the author's ranges: glucose 78 mg/dL → 4.3 mmol/L (< 5.5); WBC $14.2 \times 10^9/L$; HCO_3^- , K⁺, Na⁺ in mmol/L; blood gases in mm Hg.
- ☐ Screen 1 and screen 5 options were reordered so a correct answer is not listed first.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS9 case_1790776800009

Screen 1 · Select N (3)

Select the 3 assessment findings that are most significant.

- ✓ Temperature 101.4°F (38.5°C)
- ✓ Pulse oximetry reading 85%
- ✓ Productive cough of purulent sputum

Screen 2 · Matrix (one per row)

For each finding, click to specify if the finding is associated or not associated with COPD.

- Occupation → Associated with COPD
- Hypertension → Not Associated with COPD
- Pulse oximetry reading 85% → Associated with COPD
- 50 pack-year smoking history → Associated with COPD
- Productive cough → Associated with COPD
- Clubbing in fingers → Associated with COPD
- PaCO₂ 51 mm Hg → Associated with COPD
- Vaccine status → Not Associated with COPD

Screen 3 · Drag and drop

Drag the most appropriate word choice from the list of options to fill in the blank of the following sentence.

- The top priority for this client is to [treat the pneumonia].

Screen 4 · Matrix (one per row)

For each potential nursing or collaborative intervention, click to specify whether the intervention is appropriate or not appropriate to include in the plan of care.

- Administer IV methylprednisolone → Appropriate
- Administer PO acetaminophen as needed for fever → Appropriate
- Administer oxygen to achieve pulse oximetry readings of at least 95% → Not Appropriate
- Encourage flutter valve or Acapella use every 2 hours → Appropriate
- Administer PO cough suppressant → Not Appropriate
- Administer IV ampicillin/sulbactam → Not Appropriate
- Restrict PO fluids → Not Appropriate
- Encourage pursed-lip breathing → Appropriate
- Monitor WBC count → Appropriate
- Encourage milk, ice cream, and cheese as sources of protein → Not Appropriate

Screen 5 · Select all that apply

What should the nurse teach the client about the treatment plan before discharge? Select all that apply.

- ✓ Continue to increase fluid intake.
- ✓ Continue the flutter valve or Acapella at least 3–4 times daily.
- ✓ Obtain pneumococcal and influenza vaccines at the next provider visit.
- ✓ Eat smaller, more frequent high-calorie, high-protein meals.
- ✓ Consider strategies to stop smoking.

Screen 6 · Matrix (one per row)

For each client statement, click to specify whether the statement indicates an understanding or no understanding of the discharge teaching provided.

- “I should take the prednisone in the morning with food.” → Understanding
- “I can get back to my previous level of activity as I recover.” → Understanding
- “I can stop taking the antibiotic now that I am feeling better.” → No Understanding
- “I should use a spacer with my metered-dose inhalers.” → Understanding
- “I can continue to smoke, as the damage is already done.” → No Understanding
- “I should do pursed-lip breathing several times daily.” → Understanding
- “I should avoid crowds during cold and flu season.” → Understanding

University of Maryland - CS9 Bowtie standalone_1790776801091

Question · Bow-tie

Complete the diagram by dragging from the choices below to specify what condition the client is most likely experiencing, 2 actions the nurse should take to address that condition, and 2 parameters the nurse should monitor to assess the client's progress.

- Condition: Exacerbation of COPD
- Actions: Administer IV glucocorticoid; Administer IV antibiotic
- Parameters: Arterial blood gases; WBC count

University of Maryland - CS10: Deep Vein Thrombosis II

Source: "Deep-Vein-Thrombosis-II.docx" (medical-surgical) · Author: Dawn Leukhardt, MSN, RN, College of Southern Maryland, La Plata, Maryland · In the bank, hidden (draft)

What to review

- ☐ Screen 2: the source table's finding labels are BLANK (only the answer marks survive). The rows were rebuilt from the rationale: Edema (all 3), Decreased pedal pulse (DVT), Pain (all 3), Warmth to the calf (DVT + infection), Limited range of motion (soft tissue injury). Please confirm the labels and their order against the online version.
- ☐ A weak pedal pulse is keyed as a DVT finding (screens 1 and 2). DVT is a venous problem and does not usually reduce arterial pulses; a weak pulse suggests arterial compromise. Consider revising the chart finding and key.
- ☐ Screens 4-5: strict bedrest and "ambulation contraindicated" reflect older practice. Current guidelines (CHEST/ASH) support early ambulation once anticoagulation has started. Consider rewording (e.g. "until anticoagulation is therapeutic") or changing the key. Compression stockings for acute DVT are also no longer routinely recommended.
- ☐ Heparin orders are weight-based (80 units/kg bolus, 18 units/kg/hr), but no weight is charted. Consider adding the client's weight.
- ☐ Screen 6: "I will eat more spinach and kale every day" is keyed no understanding (source: avoid green leafy vegetables). Current teaching is to keep vitamin K intake consistent rather than avoid it; the rationale now says so.
- ☐ Screens 3-4 of the source add "Positive Homan's sign" to the admission note, but screens 1, 2, 5 and 6 do not. It was left out everywhere (Homans' sign is unreliable and no longer taught as a DVT sign), so the chart does not change between screens.
- ☐ Screen 1 options were reordered so a correct answer is not listed first. No laboratory values in this case.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS10 case_1790776800010

Screen 1 · Select N (4)

Select the 4 findings that are most concerning.

- ✓ 2+ pedal edema
- ✓ Pain 8/10 in right calf
- ✓ Warmth noted to right calf
- ✓ Right pedal pulse weak

Screen 2 · Matrix (multiple per row)

For each potential assessment finding, click to indicate if the finding is consistent with a deep vein thrombosis, a soft tissue injury, or an infection. Each finding may be consistent with more than one condition.

- Edema → Deep Vein Thrombosis, Soft Tissue Injury, Infection
- Decreased pedal pulse → Deep Vein Thrombosis
- Pain → Deep Vein Thrombosis, Soft Tissue Injury, Infection
- Warmth to the calf → Deep Vein Thrombosis, Infection
- Limited range of motion → Soft Tissue Injury

Screen 3 · Drag and drop

Drag the most appropriate choice from the list of options to fill in the blank of the following sentence.

- The nurse should recognize that the client is most likely experiencing [right leg deep vein thrombosis].

Screen 4 · Matrix (one per row)

For each potential intervention, click to specify whether the intervention is indicated or contraindicated in the plan of care.

- Obtain partial thromboplastin times → Indicated
- Begin anticoagulation therapy → Indicated
- Compression stockings to bilateral lower extremities → Indicated
- Ambulate client 3 times a day → Contraindicated
- Elevate right lower extremity → Indicated
- Massage right lower extremity 2 times daily → Contraindicated

Screen 5 · Matrix (one per row)

For each order, click to specify if the nurse should implement the order immediately, within the next hour, or before the end of the shift.

- Draw PT/PTT → Immediately
- Administer heparin 80 units/kg IV bolus → Within the Hour
- Begin heparin at 18 units/kg/hr → Within the Hour
- Redraw PTT 6 hours after start of heparin → Before the End of the Shift
- Strict bedrest → Immediately
- Place compression stockings on lower extremities → Before the End of the Shift
- Elevate right lower extremity → Within the Hour
- Closely monitor for signs of bleeding → Within the Hour

Screen 6 · Select all that apply

Which client statements indicate the teaching was understood? Select all that apply.

- ✓ "I will throw away my straight razor and buy an electric one."
- ✓ "If I miss a dose of my medicine, I will contact my doctor."
- ✓ "I set an alarm so I take my medicine at the same time daily."

University of Maryland - CS10 Bowtie standalone_1790776801101

Question · Bow-tie

Complete the diagram by dragging from the choices below to specify what condition the client is most likely experiencing, 2 actions the nurse should take to address that condition, and 2 parameters the nurse should monitor to assess the client's progress.

- Condition: Pulmonary embolism
- Actions: Place client in high-Fowler's position; Prepare client for a STAT CT scan of the chest
- Parameters: Respiratory status; Continuous cardiac monitoring

University of Maryland - CS11: End Stage Renal Disease and Dialysis

Source: "End-Stage-Renal-Disease-and-Dialysis.docx" (medical-surgical) · Author: Laura Sessions, PhD, MScN, RN, CNE, Towson University · In the bank, hidden (draft)

What to review

- ☐ Screen 2: the source's finding rows had values that differ from the lab report (creatinine 11.7 vs 10.7 mg/dL, phosphorus 5.4 vs 5.8 mg/dL, neutrophils 89% vs 89.8%); the chart's values are used, in SI units.
- ☐ Screen 4: enoxaparin keyed "not appropriate in ESRD". Dose-adjusted enoxaparin is sometimes used in severe CKD, though unfractionated heparin is preferred; consider "Enoxaparin 40 mg subcutaneous daily" wording or a clearer rationale. Vancomycin keyed not appropriate (gram-negative rods); note that empiric PD-peritonitis therapy (ISPD) covers gram-positive organisms too until cultures return.
- ☐ Screen 5 uses LPN (US term); in Ontario this is the RPN. Delegation keys follow US scope; please confirm for your program.
- ☐ Screen 6: "leave the exit site open to air or covered by loose clothing" is keyed does not understand (rationale: cover with sterile gauze). Exit-site dressing practice varies once the site is healed; please confirm with your program's PD teaching.
- ☐ Dialysate WBC reference range is "Few" in the source; the usual peritonitis threshold is > 100 cells/μL with > 50% neutrophils. Consider adding it as the reference range.
- ☐ Labs in SI with the author's ranges: urea 46 mg/dL → 16.4 mmol/L (3.6-7.1); creatinine 10.7 mg/dL → 946 μmol/L (80-124); CRP 61.5 mg/L (< 10 mg/L; source range "< 1.0 mg/dL"); albumin "3.7 g/L" (typo for 3.7 g/dL) → 37 g/L; calcium 7.9 mg/dL → 1.97 mmol/L; phosphorus 5.8 mg/dL → 1.87 mmol/L; Hgb 10.6 g/dL → 106 g/L; Hct 0.31.
- ☐ Bow-tie: "Adult respiratory distress syndrome" now reads "Acute respiratory distress syndrome".
- ☐ Screen 1 options were reordered so a correct answer is not listed first.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS11 case_1790776800011

Screen 1 · Select N (5)

Select the 5 findings that are most concerning.

- ✓ Temperature
- ✓ Periumbilical tenderness
- ✓ Peritoneal dialysis catheter exit site
- ✓ Serum WBC count
- ✓ Dialysate WBC count

Screen 2 · Matrix (multiple per row)

For each client finding, click to specify if the finding supports the diagnosis of peritonitis or end-stage renal disease (ESRD). Each finding may support more than one condition.

- Urea 16.4 mmol/L → End-Stage Renal Disease (ESRD)
- Creatinine 946 μmol/L → End-Stage Renal Disease (ESRD)
- C-reactive protein 61.5 mg/L → Peritonitis
- Potassium 5.86 mmol/L → End-Stage Renal Disease (ESRD)
- Phosphate 1.87 mmol/L → End-Stage Renal Disease (ESRD)
- Hemoglobin 106 g/L → End-Stage Renal Disease (ESRD)
- WBC $14.22 \times 10^9/L$ → Peritonitis
- Neutrophils 89.8% → Peritonitis

Screen 3 · Drag and drop

Drag the most appropriate choice from the list of options to fill in the blank of the following sentence.

- The client is at highest risk for developing [sepsis].

Screen 4 · Matrix (one per row)

Based on the new laboratory data, for each potential order, click to specify whether the order is appropriate or not appropriate to include in the plan of care.

- Daily weights → Appropriate
- Blood cultures STAT → Appropriate
- Peritoneal dialysis catheter care every shift → Appropriate
- IV D5 ½ NS with 20 mmol/L KCl at 120 mL/hr → Not Appropriate
- Calcium carbonate with meals → Appropriate
- Vancomycin IV → Not Appropriate
- Gentamicin intraperitoneally daily → Appropriate
- Enoxaparin subcutaneous daily → Not Appropriate
- Epoetin alfa injections → Appropriate

Screen 5 · Matrix (one per row)

For each task, click to specify if the task should be performed by the RN, the LPN, or the UAP.

- Vital signs → UAP
- Dialysate exchange → RN
- Daily weights → UAP
- Peritoneal dialysis catheter care every shift → LPN
- Intraperitoneal gentamicin → RN
- Peritoneal dialysis catheter care teaching → RN

Screen 6 · Matrix (one per row)

For each client statement, click to specify whether the statement indicates the client understands or does not understand the teaching about self-care of the peritoneal dialysis catheter.

- “Before cleaning the area, I will wash my hands with soap and water and put on clean gloves.” → Understands
- “I should remove crusts or scabs at the exit site before washing the site.” → Does Not Understand
- “I should hold the catheter in place during cleaning to prevent injury to the skin.” → Understands
- “I should scrub the exit site vigorously with an antiseptic solution like iodine or chlorhexidine.” → Does Not Understand
- “I will put antibiotic cream on the skin around the catheter with a cotton-tip swab every time I change the dressing.” → Understands
- “I will not use any creams with petroleum because they can damage the catheter.” → Understands
- “I will leave the exit site open to air or covered by loose clothing.” → Does Not Understand

University of Maryland - CS11 Bowtie standalone_1790776801111

Question · Bow-tie

Based on the client information, complete the diagram by dragging from the choices below to specify what condition the client is most likely experiencing, 2 actions the nurse should take to address that condition, and 2 parameters the nurse should monitor to assess the client's progress.

- Condition: Septicemia
- Actions: Administer oxygen; Administer broad-spectrum antibiotics IV
- Parameters: Pulse oximetry reading; Blood pressure

University of Maryland - CS12: Gastroesophageal Reflux Disease

Source: "Gastroesophageal-Reflux.docx" (medical-surgical) · Author: Tara Sohrabi, Nursing Professor, Montgomery College · In the bank, hidden (draft)

What to review

- ☐ Chest pain: a 47-year-old with intermittent chest pain and epigastric pain would normally have an ECG and troponin to rule out acute coronary syndrome before GERD is diagnosed. Screen 1 keys chest pain as not needing immediate follow-up. Consider adding "ECG normal sinus rhythm; troponin negative" to the chart so the reasoning holds.
- ☐ Orders: no acid suppression (PPI/H2 blocker or antacid) is ordered for probable GERD, morphine is the analgesic (opioids are not typical for GERD pain), and potassium 3.0 mmol/L has no replacement order. Please review the order set.
- ☐ Screen 3 rationale mentioned "difficulty swallowing", which is not in the chart; replaced with the charted findings.
- ☐ Screen 5: "Sleep on your right side" keyed incorrect; a sentence was added explaining that the left side is preferred (the source only mentioned elevating the head of the bed).
- ☐ Laboratory report: the source's sodium reference range reads "145 mEq/L"; shown as 135-145 mmol/L.
- ☐ Screen 1 is a highlight question (chart passage and question in the left panel); the key (epigastric pain; nausea and vomiting) comes from the source's highlighted Key passage. Students may select 2.
- ☐ Screen 5 options were reordered so a correct answer is not listed first.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS12 case_1790776800012

Screen 1 · Highlight (passage left)

Click to highlight the 2 findings that require immediate follow-up.

- ✓ epigastric pain
- ✓ nausea and vomiting

Screen 2 · Matrix (multiple per row)

For each finding, click to specify if the finding is most consistent with gastroesophageal reflux disease, peptic ulcer, or cholecystitis. Each finding may support more than one condition.

- Epigastric pain → Gastroesophageal Reflux Disease, Peptic Ulcer, Cholecystitis
- Intermittent chest pain → Gastroesophageal Reflux Disease, Cholecystitis
- Vomiting → Gastroesophageal Reflux Disease, Peptic Ulcer, Cholecystitis
- Cough → Gastroesophageal Reflux Disease

Screen 3 · Drop-down cloze

Complete the following sentence by choosing from the list of options.

- The nurse should recognize that the client is most likely experiencing [gastroesophageal reflux disease].

Screen 4 · Select all that apply

Which interventions should the nurse include in the plan of care? Select all that apply.

- ✓ Check lab report for sodium and potassium levels
- ✓ Administer antiemetics
- ✓ Teach the client about health maintenance
- ✓ Administer analgesics

Screen 5 · Select all that apply

Which modifications should the nurse include in the teaching plan? Select all that apply.

- ✓ Avoid eating large meals
- ✓ Avoid eating late at night
- ✓ Eat a low-fat diet
- ✓ Limit caffeinated beverages
- ✓ Maintain a healthy weight

Screen 6 · Multiple choice

The nurse determines that the dietary teaching was successful if the client chooses which food?

- ✓ Nonfat milk

University of Maryland - CS12 Bowtie standalone_1790776801121

Question · Bow-tie

Complete the diagram by dragging from the choices below to specify what condition the client is most likely experiencing, 2 actions the nurse should take to address that condition, and 2 parameters the nurse should monitor to assess the client's progress.

- Condition: Gastroesophageal reflux disease
- Actions: Administer analgesics; Administer antiemetic
- Parameters: Nausea and vomiting; Electrolyte levels

University of Maryland - CS13: Heart Failure

Source: "Heart-Failure.docx" (medical-surgical) · Author: Lisa Seldomridge, PhD, RN, CNE; Jennifer Hart, DNP, FNP-BC; Molly Dale, DNP, FNP-BC; Salisbury University · In the bank, hidden (draft)

What to review

- ☐ Home medications: hydrochlorothiazide 25 mg PO TWICE a day (usually once daily; a loop diuretic is more typical for HF with this degree of fluid overload). Please review.
- ☐ Screen 5 key: questioning sacubitril/valsartan with enalapril is correct (ARNI + ACE inhibitor contraindicated; 36-hour washout). The rationale now states the reason (angioedema risk) and why an extra diuretic dose and starting today are wrong.
- ☐ Screen 3 rationale said the creatinine was "within normal limits"; it is above the range (1.5 mg/dL = 133 µmol/L vs 80-124). Rationale corrected to "slightly elevated".
- ☐ Labs: Hct 34.6% with Hgb 13.5 g/dL is physiologically inconsistent (Hgb normal, Hct low). Consider Hgb ~11.5 g/dL (115 g/L) or Hct ~40%.
- ☐ Screen 2 was a two-blank drag-and-drop where either order is correct; the app scores blanks by position, so it is a Select 2 question here (same options and key).
- ☐ Screen 4: weight units shown in lb and kg; a sentence was added giving the usual reporting threshold (2-3 lb/day or 5 lb/week) to explain why the 1-lb option is keyed incorrect.
- ☐ Screen 6: thirst is keyed Unchanged although the Day 1 note does not mention thirst; consider rewording the row.
- ☐ Labs in SI with the author's ranges: cholesterol 248 mg/dL → 6.4 mmol/L; BUN 10 mg/dL → urea 3.6 mmol/L; creatinine 1.5 mg/dL → 133 µmol/L; Hct 0.346 L/L; Hgb 135 g/L; K+/Na+/Cl- mmol/L.
- ☐ Screen 3, 4 and 5 options were reordered so a correct answer is not listed first.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS13 case_1790776800013

Screen 1 · Select N (4)

Select the 4 findings that are most concerning.

- ✓ 4+ pitting pedal and pretibial edema
- ✓ Crackles in bilateral lung bases
- ✓ Awakening at night with shortness of breath
- ✓ Weight gain

Screen 2 · Select N (2)

Select the 2 problems the client is at most risk for developing.

- ✓ Electrolyte imbalance
- ✓ Fluid volume overload

Screen 3 · Multiple choice

Which abnormal laboratory finding should the nurse address first?

- ✓ Potassium

Screen 4 · Select all that apply

What should the nurse include in the teaching plan for this client? Select all that apply.

- ✓ Weigh yourself every day at the same time and in the same clothes.
- ✓ Limit sodium in the diet to no more than 2 grams per day.
- ✓ Take prescribed medication every day, even if feeling better.
- ✓ Eat foods high in potassium.

Screen 5 · Select all that apply

Which actions should the nurse take in response to the change in the medical orders? Select all that apply.

- ✓ Question the new medication order
- ✓ Educate client about stopping digoxin
- ✓ Request that the provider order a potassium supplement
- ✓ Ask the dietitian to meet with the client
- ✓ Determine if the client owns a scale

Screen 6 · Matrix (one per row)

For each finding, click to specify if the finding indicates that the client's status has improved, worsened, or is unchanged from his condition at the clinic visit 2 weeks earlier.

- Sleeping in recliner → Worsened
- Nagging cough → Worsened
- Pink-tinged sputum → Worsened
- Difficulty catching breath → Worsened
- Pulse → Improved
- Lack of appetite → Worsened
- Getting up at night to urinate → Worsened
- Thirst → Unchanged

University of Maryland - CS13 Bowtie standalone_1790776801131

Question · Bow-tie

Complete the diagram by dragging from the choices below to specify what condition the client is most likely experiencing, 2 actions the nurse should take to address that condition, and 2 parameters the nurse should monitor to assess the client's progress.

- Condition: Heart failure
- Actions: Administer IV diuretic; Place client on fluid restriction
- Parameters: Weight; Lung sounds

University of Maryland - CS14: HIV with an Opportunistic Infection

Source: "HIV-with-Opportunistic-Infection.docx" (medical-surgical) · Author: Denyce Watties-Daniels, DNP, RN, Coppin State University · In the bank, hidden (draft)

What to review

- ☐ Screen 4 keys "Administer ibuprofen as needed for fever" as anticipated; the client's creatinine is elevated (1.5 mg/dL = 133 µmol/L), where NSAIDs are usually avoided (acetaminophen preferred). Consider changing the order to acetaminophen.
- ☐ PCP with hypoxemia (SpO2 90% on room air): adjunctive corticosteroids are usually given with TMP-SMX when PaO2 < 70 mm Hg. Not part of any question; consider adding to the orders/rationale.
- ☐ Screen 1: sodium 148 mmol/L (above range) is not keyed as concerning. Consider keying it or changing the value.
- ☐ Screen 6 keys "I can stop wearing a mask when I feel better" as no understanding; masks are not a standard precaution for PCP itself. The rationale now frames it as protection while the CD4 count is low; please confirm.
- ☐ Screen 6: the source rationale was generic; it now explains each statement.
- ☐ Source rationale said the normal CD4 count is 800-1200 cells/mm3 while the lab report range is 500-1600; the rationale no longer quotes a range.
- ☐ Labs in SI with the author's ranges: BUN 22 mg/dL → urea 7.9 mmol/L; creatinine 1.5 mg/dL → 133 µmol/L; Hct 0.48 L/L; Hgb 150 g/L; WBC 15.2 × 10⁹/L (source "15.2 mm3"); platelets 300 × 10⁹/L; CD4 kept in cells/µL (as MCC prints it).
- ☐ Screen 5 options were reordered so a correct answer is not listed first.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS14 case_1790776800014

Screen 1 · Select all that apply

Which laboratory findings are concerning? Select all that apply.

- ✓ CD4 T-cell count
- ✓ WBC count

Screen 2 · Select all that apply

Which client findings are risks for an opportunistic infection? Select all that apply.

- ✓ History of HIV
- ✓ Stopped taking his medication
- ✓ Low CD4 T-cell count

Screen 3 · Dyad

Complete the following sentences by choosing from the lists of options.

- The nurse should first address the client's [oxygenation]. Next, the nurse should address the client's [infection].

Screen 4 · Matrix (one per row)

For each potential order, click to specify if the order is anticipated or not anticipated in the plan of care.

- Titrate oxygen to keep pulse oximetry reading above 95% → Anticipated
- Administer ibuprofen as needed for fever → Anticipated
- Administer trimethoprim/sulfamethoxazole → Anticipated
- Administer vancomycin → Not Anticipated
- Gargle with lidocaine solution → Not Anticipated
- Schedule a stress test in the morning → Not Anticipated
- Refer to social work → Anticipated

Screen 5 · Select all that apply

What should the nurse teach the client about preventing opportunistic infections? Select all that apply.

- ✓ Wash your hands frequently.
- ✓ Get tested for tuberculosis.
- ✓ Get tested for sexually transmitted infections.
- ✓ Get a yearly influenza vaccine.
- ✓ Do not drink untreated water.
- ✓ Have viral load testing every 4 to 6 months.
- ✓ Take antiretroviral medications as prescribed.

Screen 6 · Matrix (one per row)

For each client statement, click to specify if the statement indicates understanding or no understanding of the teaching provided.

- “My low CD4 count shows I am at higher risk for infection.” → Understanding
- “I should avoid crowds when possible.” → Understanding
- “It is a good practice to keep hand sanitizer in my car and my pocket.” → Understanding
- “I can stop wearing a mask when I feel better.” → No Understanding
- “Washing my hands frequently is one of the best ways to prevent infection.” → Understanding
- “I need to keep using safe sex practices.” → Understanding
- “I will schedule an HIV test every 6 months.” → No Understanding
- “I will stay current with my vaccinations.” → Understanding
- “Eating a balanced diet and maintaining good nutrition will help improve my immune system.” → Understanding
- “Not taking a full course of antibiotics could put me at risk for future drug-resistant infections.” → Understanding

University of Maryland - CS14 Trend standalone_1790776801141

Question · Matrix (one per row)

For each client finding, click to indicate if the finding reflects that the client's condition has improved or is unchanged.

- Temperature → Improved
- Cough → Improved
- Pulse → Improved
- Dyspnea → Improved
- Blood pressure → Unchanged
- Pulse oximetry reading → Improved
- Pain → Improved

University of Maryland - CS15: Type II Diabetes and Ketoacidosis

Source: "Ketoacidosis.docx" (medical-surgical) · Author: DeNiece Bennett, DNP, MSN-Ed, University of Maryland School of Nursing · In the bank, hidden (draft)

What to review

- ☐ CORRECTED blood gas: the source's ABG (pH 7.20, PaCO₂ 45, HCO₃ 32) is not DKA (a high bicarbonate means metabolic alkalosis/compensated respiratory acidosis). Changed to PaCO₂ 25 mm Hg and HCO₃ 10 mmol/L (metabolic acidosis with respiratory compensation; still pH 7.20). Same change in the bow-tie. Please confirm.
- ☐ CORRECTED potassium order: "0.9 NS with 20 mEq KCl/100 mL" (= 200 mmol/L, dangerous) is now "0.9% sodium chloride with 20 mmol/L KCl" (20 mmol per litre). Please confirm the intended concentration.
- ☐ CHANGED KEY (screen 4): "Teach the client pursed-lip breathing" was keyed Indicated; it is now Contraindicated, because the Kussmaul respirations compensate for the metabolic acidosis. Please confirm, or replace the row.
- ☐ Screen 4 also keys insulin glargine SC and an indwelling catheter as Contraindicated; some DKA protocols give early basal insulin, and ICU care may use a catheter for hourly output. Consider "Not indicated at this time".
- ☐ Vital signs: the source's last two columns are both labelled 1330; they are 1300 and 1330 here (matching the notes).
- ☐ The 1215 note on screen 2 includes "2 L O₂ started per NC" but screens 3-6 omit it; it is kept on every screen from screen 2 so the chart does not lose it.
- ☐ SpO₂ 89% on room air with clear lungs and a PaO₂ of 60 (bow-tie) is unusual for DKA alone; consider whether it is intended.
- ☐ Home medication: sitagliptin/metformin 50/1000 mg once daily (usually twice daily). Empagliflozin (an SGLT2 inhibitor) is a useful cue for DKA in type 2 diabetes; consider mentioning it in a rationale.
- ☐ Labs in SI with the author's ranges: glucose 435-440 mg/dL → 24.2-24.4 mmol/L (3.9-7.8); creatinine 1.9 mg/dL → 168 µmol/L; BUN 34 mg/dL → urea 12.1 mmol/L; K⁺/Na⁺ mmol/L; Hct 0.37; Hgb 140 g/L. The screen 2 rationale's glucose ranges were converted (300-800 mg/dL ≈ 14-44 mmol/L; > 600 mg/dL ≈ > 33 mmol/L).
- ☐ Screen 6 options were reordered so a correct answer is not listed first.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS15 case_1790776800015

Screen 1 · Select N (4)

Select the 4 findings that are most significant.

- ✓ History of type 2 diabetes
- ✓ Respiratory status
- ✓ Circulation
- ✓ Mental status

Screen 2 · Matrix (multiple per row)

For each finding, click to specify if it is consistent with ketoacidosis or hyperglycemic hyperosmolar syndrome. Each finding may support more than one condition.

- pH 7.20 → Ketoacidosis
- Blood glucose 24.2 mmol/L → Ketoacidosis
- Serum creatinine 168 µmol/L → Ketoacidosis, Hyperglycemic Hyperosmolar Syndrome
- Urine ketones positive → Ketoacidosis

Screen 3 · Drop-down cloze

Complete the following sentence by choosing from the list of options.

- The problem the nurse should address first is [restoring volume].

Screen 4 · Matrix (one per row)

For each nursing intervention, click to specify whether the intervention is indicated, contraindicated, or nonessential.

- Obtain glycosylated hemoglobin (A1C) → Nonessential
- Teach the client pursed-lip breathing → Contraindicated
- Administer insulin glargine subcutaneously → Contraindicated
- Insert indwelling urinary catheter → Contraindicated
- Administer IV potassium → Indicated
- Monitor ECG → Indicated

Screen 5 · Select all that apply

What actions should the nurse take while implementing the treatment plan? Select all that apply.

- ✓ Anticipate holding insulin for low potassium levels
- ✓ Request IV fluids with dextrose when glucose levels start to normalize
- ✓ Allow the client to eat when status stabilizes
- ✓ Monitor for symptoms of fluid overload

Screen 6 · Select all that apply

Which findings indicate the treatment plan has been effective? Select all that apply.

- ✓ Blood pressure
- ✓ Mental status
- ✓ Pulse 100

University of Maryland - CS15 Bowtie standalone_1790776801151

Question · Bow-tie

Complete the diagram by dragging from the choices below to specify what condition the client is most likely experiencing, 2 actions the nurse should take to address that condition, and 2 parameters the nurse should monitor to assess the client's progress.

- Condition: Diabetic ketoacidosis
- Actions: Administer IV insulin; Infuse normal saline fluid bolus
- Parameters: Capillary glucose; Level of consciousness

University of Maryland - CS16: Liver Failure

Source: "Liver-Failure.docx" (medical-surgical) · Author: Mary DiBartolo, PhD, RN-BC, CNE, FGSA, FAAN, Salisbury University · In the bank, hidden (draft)

What to review

- ☐ Screen 4 keys "acetaminophen 650 mg PRN for fever" as CONTRAINDICATED. Current guidance (e.g. AASLD) considers acetaminophen up to about 2 g/day the preferred analgesic/antipyretic in cirrhosis (NSAIDs are the drugs to avoid). Consider changing the key to Indicated, or the option to an NSAID (e.g. ibuprofen).
- ☐ Screen 4: point-of-care glucose is keyed Not indicated, but this malnourished client with alcohol use disorder has a fasting glucose of 70 mg/dL (3.9 mmol/L) and is at risk of hypoglycemia. Please confirm.
- ☐ Screen 1 rationale said dyspnea indicates "hepatopulmonary syndrome from worsening ascites" and that the creatinine is normal (no creatinine is charted). Rewritten: dyspnea from ascites limiting diaphragm movement; the creatinine sentence was removed. Consider adding a creatinine value to the lab report.
- ☐ Screen 5 is a highlight question (question on the right, the order list as the passage; chart on the left). Its key (oxygen, lactulose, IV furosemide) is from the source's highlighted Key; students may select 3.
- ☐ Screen 6 rationale now explains the second blank (explore readiness to stop drinking) and why the other choices are wrong.
- ☐ Medication list: metoprolol XL 50 mg daily (a selective beta-blocker; nonselective beta-blockers are used for portal hypertension). Not tested; consider whether intended.
- ☐ Labs in SI with the author's ranges: BUN 22 mg/dL → urea 7.9 mmol/L; glucose 70 mg/dL → 3.9 mmol/L; ammonia 94 mcg/dL → 55 µmol/L (6-47) and Day 2 72 mcg/dL → 42 µmol/L (also added to the lab table on screen 6); albumin 3.2 g/dL → 32 g/L.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS16 case_1790776800016

Screen 1 · Select N (2)

Select the 2 findings that are most concerning.

- ✓ Neurologic assessment
- ✓ Dyspnea

Screen 2 · Matrix (one per row)

For each finding, click to specify if the finding is most consistent with ascites or an elevated ammonia level.

- Abdominal girth → Ascites
- Agitation → Elevated Ammonia
- Bulging flanks → Ascites
- Dozing off and on → Elevated Ammonia
- Dyspnea → Ascites
- Peripheral edema → Ascites
- Pulse oximetry reading → Ascites
- Serum albumin → Ascites

Screen 3 · Dyad

Complete the following sentence by choosing from the lists of options.

- The client is most likely experiencing [hepatic encephalopathy], as evidenced by the [neurologic assessment].

Screen 4 · Matrix (one per row)

For each potential intervention, click to specify whether the intervention is indicated, not indicated, or contraindicated in the plan of care.

- Neuro assessment every 4 hours → Indicated
- Record abdominal girth daily → Indicated
- Administer acetaminophen 650 mg as needed for temperature above 101°F (38.3°C) → Contraindicated
- Monitor pulse oximetry every 4 hours → Indicated
- Administer sedative as needed for agitation → Contraindicated
- Daily weights → Indicated
- Supine position when in bed → Contraindicated
- Low-sodium diet → Indicated
- Point-of-care glucose every 6 hours → Not Indicated
- Lactulose 30 mL every 4 hours × 3 doses → Indicated

Screen 5 · Highlight

Click to highlight the 3 orders that the nurse should implement immediately.

- ✓ Oxygen per nasal cannula to maintain pulse oximetry reading at 95% or greater
- ✓ Lactulose 30 mL PO every 4 hours × 3 doses
- ✓ Furosemide 20 mg IV

Screen 6 · Dyad

Complete the following sentences by choosing from the lists of options.

- The nurse determines the client's status is [improving]. The nurse should now [explore readiness to stop drinking].

University of Maryland - CS16 Bowtie standalone_1790776801161

Question · Bow-tie

Complete the diagram by dragging from the choices below to specify what condition the client is most likely experiencing, 2 actions the nurse should take to address that condition, and 2 parameters the nurse should monitor to assess the client's progress.

- Condition: Hepatic encephalopathy
- Actions: Administer lactulose; Institute safety precautions
- Parameters: Neurologic status; Serum ammonia

University of Maryland - CS17: Prostate Cancer with Prostatectomy

Source: "Prostate-Cancer.docx" (medical-surgical) · Author: Tara Sohrabi, Nursing Professor, Montgomery College · In the bank, hidden (draft)

What to review

- ☐ Screen 3 has NO answer key in the source. Keyed as hemorrhagic shock + "cardiovascular assessment" (the rationale cites low BP, tachycardia, and bleeding). Please confirm the second blank.
- ☐ Ketorolac 30 mg IV is given at 1345 to a client who is bleeding after surgery (NSAIDs increase bleeding risk). Not tested; consider replacing it with an opioid or using it as a cue in a rationale.
- ☐ Continuous bladder irrigation with catheter traction is standard after TURP, but uncommon after a robotic-assisted RADICAL prostatectomy (which has a urethrovesical anastomosis). Consider changing the procedure to TURP, or removing CBI/traction.
- ☐ Screen 6 vital signs in the source show RR 14 at 1345 and 1415, while screens 1-5 show 20; the earlier values (20, 20) are used throughout so the chart does not change.
- ☐ Screen 5 is a highlight question (question on the right, the order list as the passage; chart on the left); its key (fluid bolus, oxygen) is from the source's highlighted Key. "Maintain traction on catheter" is not selectable (already done by the physician).
- ☐ Screen 4 stem typo fixed ("appropriate or not appropriate").
- ☐ Labs in SI with the author's ranges: Hct 32% → 0.32 L/L; Hgb 10 g/dL → 100 g/L; transfusion threshold Hgb < 7 g/dL → < 70 g/L.
- ☐ Screen 1 options were reordered so a correct answer is not listed first.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS17 case_1790776800017

Screen 1 · Select all that apply

Which findings require immediate follow-up? Select all that apply.

- ✓ Mental status
- ✓ Heart rate
- ✓ Blood pressure
- ✓ Pulse oximetry reading
- ✓ Urine

Screen 2 · Matrix (multiple per row)

For each finding, click to specify if the finding is consistent with the complication of hemorrhagic shock, fluid overload, or pulmonary embolism. Each finding may support more than one complication.

- Confusion → Hemorrhagic Shock, Pulmonary Embolism
- Rapid heart rate → Hemorrhagic Shock, Pulmonary Embolism
- Low blood pressure → Hemorrhagic Shock, Pulmonary Embolism
- Low pulse oximetry reading → Hemorrhagic Shock, Fluid Overload, Pulmonary Embolism
- Bright red urine → Hemorrhagic Shock

Screen 3 · Dyad

Complete the following sentence by choosing from the lists of options.

- The client is most likely experiencing [hemorrhagic shock], as evidenced by the [cardiovascular assessment].

Screen 4 · Matrix (one per row)

For each potential intervention, click to specify whether the intervention is appropriate or not appropriate to include in the plan of care for a client experiencing hemorrhagic shock after prostatectomy.

- Obtaining a hemoglobin and hematocrit → Appropriate
- Removing the urinary catheter → Not Appropriate
- Administering a fluid bolus → Appropriate
- Obtaining a urine culture → Not Appropriate
- Administering an enema → Not Appropriate
- Assessing urine output hourly → Appropriate
- Applying oxygen → Appropriate

Screen 5 · Highlight

Click to highlight 2 additional orders the nurse should implement immediately to manage hemorrhagic shock.

- ✓ Administer 500 mL 0.9% sodium chloride IV bolus
- ✓ Administer oxygen to keep oxygen saturation above 94%

Screen 6 · Dyad

Complete the following sentences by choosing from the lists of options.

- The nurse determines the client's status is [improving]. The nurse should now [continue monitoring the client's vital signs].

University of Maryland - CS17 Bowtie standalone_1790776801171

Question · Bow-tie

Complete the diagram by dragging from the choices below to specify what condition the client is most likely experiencing, 2 actions the nurse should take to address that condition, and 2 parameters the nurse should monitor to assess the client's progress.

- Condition: Hemorrhagic shock
- Actions: Administer oxygen; Administer IV fluid bolus
- Parameters: Blood pressure; Urine

University of Maryland - CS18: Care of Client Post-Op Spine Surgery (Laminectomy)

Source: "Spine-Surgery.docx" (medical-surgical) · Author: Mary DiBartolo, PhD, RN-BC, CNE, FAAN, and Allison Hynson, MSN, RN · In the bank, hidden (draft)

What to review

- ☐ Nurses' notes: pupils were recorded as "3 cm"; changed to 3 mm.
- ☐ Screen 1 listed "Urinary retention" twice; it appears once.
- ☐ Bow-tie: a surgical site infection on postoperative day 1-2 is early (SSIs usually appear after day 3-5; early fever is more often atelectasis or inflammation). Consider moving the bow-tie to day 4-5. The source's 10-mg oxycodone line was missing its dose on the bow-tie; filled in as 10 mg.
- ☐ Discharge teaching mixes "stool softener" (screen 5-6) with the senna order (a stimulant laxative). Consider docusate, or calling senna a laxative.
- ☐ Screen 5 key: "Check the surgical dressing in the mirror daily" is correct; consider whether a family member should check it (hard to see a lumbar incision in a mirror).
- ☐ "Steri-strips" (a brand name) now read "adhesive strips".
- ☐ Labs in SI with the author's ranges: glucose 130 mg/dL → 7.2 mmol/L; Hct 0.30 L/L; Hgb 110 g/L; WBC $10.0 \times 10^9/L$ (bow-tie 17.0); platelets $200 \times 10^9/L$; HbA1c 6.0% (< 5.7%, US range; see CS2 note).
- ☐ Screen 5 options were reordered so a correct answer is not listed first.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS18 case_1790776800018

Screen 1 · Select N (4)

Select the 4 findings that require immediate follow-up.

- ✓ Jackson-Pratt drainage
- ✓ Status of surgical dressing
- ✓ Client report of bending to get call bell
- ✓ Status of Jackson-Pratt drain

Screen 2 · Select all that apply

Which laboratory value trends would the nurse expect to see with this client's condition? Select all that apply.

- ✓ Decrease in hemoglobin
- ✓ Decrease in hematocrit

Screen 3 · Drag and drop

Drag the most appropriate choice from the list of options to fill in the blank of the following sentence.

- The top priority for this client is [surgical site management].

Screen 4 · Matrix (one per row)

For each potential nursing or collaborative intervention, click to specify whether the intervention is appropriate or not appropriate to include in the plan of care.

- Notify provider → Appropriate
- Administer oxycodone 10 mg → Not Appropriate
- Compress JP drain → Appropriate
- Blood draw for hemoglobin and hematocrit → Appropriate
- Insert indwelling urinary catheter → Not Appropriate
- Administer senna 1 tablet PO twice daily → Appropriate

Screen 5 · Select all that apply

What should the nurse teach the client before discharge? Select all that apply.

- ✓ Take a prescribed stool softener daily after discharge to prevent straining.
- ✓ Do not bend, twist, or lift more than 5–10 lb (2–4.5 kg) after surgery.
- ✓ Call the provider's office if you experience fever or chills.
- ✓ Check the surgical dressing in the mirror daily.

Screen 6 · Matrix (one per row)

For each client statement, click to specify whether the statement indicates an understanding or no understanding of the teaching provided.

- "I should take the stool softener only if I feel constipated." → No Understanding
- "I should walk as much as possible after surgery." → Understanding
- "I am able to get my bandage wet and can soak in the tub to relieve pain." → No Understanding
- "I should only take the opioid medication when I experience moderate or severe pain." → Understanding
- "I should leave the adhesive strips on the incision until they fall off." → Understanding
- "I should be able to pick up my dog as long as it does not cause pain." → No Understanding
- "I should wear the brace to sleep but can take it off when I'm walking around." → No Understanding
- "I do not need to worry about the surgical dressing until I go to the provider in 2 weeks." → No Understanding

University of Maryland - CS18 Bowtie standalone_1790776801181

Question · Bow-tie

Complete the diagram by dragging from the choices below to specify what condition the client is most likely experiencing, 2 actions the nurse should take to address that condition, and 2 parameters the nurse should monitor to assess the client's progress.

- Condition: Surgical site infection
- Actions: Administer acetaminophen; Assess surgical site
- Parameters: Temperature; WBC count

University of Maryland - CS19: Stroke

Source: "Stroke.docx" (medical-surgical) · Author: DeNiece Bennett, DNP, MSN-Ed, RN, University of Maryland School of Nursing · In the bank, hidden (draft)

What to review

- ☐ Screen 1 keys "heart rate 99" as needing immediate follow-up (rationale: elevated heart rate), but 99 is within normal limits. Consider changing the value (e.g. 112) or removing it from the key.
- ☐ Head CT: the source says the CT "shows ischemic stroke". Early CT is usually normal in ischemic stroke and is done to exclude hemorrhage; the report now reads "findings consistent with acute ischemic stroke; no intracranial hemorrhage". Please confirm.
- ☐ Timing: symptoms began at 0900 and alteplase is started at 1200 (3 hours). The source rationale says "within 3 hours"; current guidelines allow up to 4.5 hours for eligible clients. The rationale now says "within the treatment window".
- ☐ Discharge on apixaban (an anticoagulant) after an ischemic stroke without atrial fibrillation is unusual (antiplatelet therapy is standard). Please review the discharge medication.
- ☐ Screen 6 keys "The physical therapist will determine my functional ability" as no understanding (source: the nurse completes the baseline functional assessment). PT/OT also assess function after stroke; consider rewording.
- ☐ Bow-tie keys supplemental oxygen as a correct action, but no SpO2 is charted; guidelines give oxygen only if SpO2 < 94%. Consider adding a low SpO2 to the chart.
- ☐ Labs: PT 13.5 s (prolonged) with no anticoagulant, Hct 55% (high for a woman) and platelets $500 \times 10^9/L$ are not explained; consider whether intended (e.g. dehydration/polycythemia).
- ☐ Labs in SI with the author's ranges: glucose 186 mg/dL → 10.3 mmol/L; alteplase glucose cut-off 50 mg/dL → 2.8 mmol/L; post-stroke glucose target 140-180 mg/dL → 7.8-10 mmol/L; Hct 0.55 L/L; Hgb 130 g/L; platelets $500 \times 10^9/L$.
- ☐ Screen 1 is a highlight question (passage and question in the left panel); its key is from the source's highlighted Key passage (5 findings). "Students may select" has no limit, as in the source.
- ☐ Screen 4 options were reordered so a correct answer is not listed first.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS19 case_1790776800019

Screen 1 · Highlight (passage left)

Click to highlight the findings that require immediate follow-up.

- ✓ lightheadedness
- ✓ generalized weakness of the left lower and upper extremities
- ✓ sudden headache
- ✓ Heart rate 99
- ✓ blood pressure 160/100

Screen 2 · Matrix (one per row)

For each finding, click to specify if the finding is a risk factor or not a risk factor for ischemic stroke.

- Combined hormonal contraceptive → Risk Factor
- Drinks 3–4 glasses of alcohol a week → Not a Risk Factor
- Vapes e-cigarettes daily → Risk Factor
- Completed a 2.5-mile sprint on the treadmill → Not a Risk Factor
- Latin descent → Risk Factor

Screen 3 · Drag and drop

Drag the most appropriate choice from the list of options to fill in the blank of the following sentence.

- The top priority for this client is [restoring cerebral perfusion].

Screen 4 · Select all that apply

What additional assessment data should the nurse obtain before administering alteplase? Select all that apply.

- ✓ No current pregnancy
- ✓ Blood glucose greater than 2.8 mmol/L
- ✓ No major surgical procedures within 14 days
- ✓ Confirm onset and time frame of symptoms
- ✓ Client weight

Screen 5 · Matrix (one per row)

For each possible action, click to specify if it is indicated or not indicated after administering alteplase.

- Ensure oral anticoagulants are withheld for the next 24 hours → Indicated
- Arrange for an inpatient admission to the medical-surgical unit → Not Indicated
- Assess and monitor vital signs for intracerebral hemorrhage → Indicated
- Insert an indwelling catheter to record urine output accurately → Not Indicated
- Consult with a hematologist to increase the client's PT and INR levels → Not Indicated
- Keep the client NPO until a speech and swallow evaluation is completed → Indicated
- Monitor serum glucose level to prevent complications of a stroke → Indicated

Screen 6 · Matrix (one per row)

For each client statement, click to specify whether the statement indicates an understanding or no understanding of the teaching provided.

- "I will use the call bell and wait for someone to help me get out of bed." → Understanding
- "A speech and swallow test will determine if I can drink normally again." → Understanding
- "I will switch to an electric razor instead of a straight razor to shave." → Understanding
- "The physical therapist will determine my functional ability." → No Understanding
- "At discharge, I should plan to self-administer insulin for hyperglycemia." → No Understanding
- "I will follow up with my gynecologist to determine a better contraceptive plan." → Understanding

University of Maryland - CS19 Bowtie standalone_1790776801191

Question · Bow-tie

Complete the diagram by dragging from the choices below to specify what condition the client is most likely experiencing, 2 actions the nurse should take to address that condition, and 2 parameters the nurse should monitor to assess the client's progress.

- Condition: Ischemic stroke
- Actions: Give supplemental oxygen; Initiate code stroke
- Parameters: Level of consciousness; Patency of airway

University of Maryland - CS20: Tension Pneumothorax

Source: "Tension-Pneumothorax.docx" (medical-surgical) · Author: Denyce Watties-Daniels, DNP, RN, Coppin State University · In the bank, hidden (draft)

What to review

- ☐ Screen 4 option "Assist with thoracentesis" changed to "Assist with needle decompression" (a tension pneumothorax is relieved by needle decompression/thoracostomy, not thoracentesis, which drains fluid). Rationale and the 1545 note reworded to match ("14-gauge spinal needle to aspirate air" → "14-gauge needle to decompress the chest").
- ☐ Screen 4 keys "Obtain a STAT chest X-ray" as anticipated and the bow-tie keys it as an action. Tension pneumothorax is a clinical diagnosis and imaging must not delay decompression; the rationales say so. Consider replacing it in the bow-tie.
- ☐ Screen 4 keys an indwelling urinary catheter as not anticipated; in a hypotensive, stuporous trauma client in the ICU one is often placed for output monitoring. Please confirm.
- ☐ Screen 2 keys "Muffled heart sounds" for tension pneumothorax as well as tamponade (classically a tamponade sign, Beck's triad). The row "Blood gas" was renamed "Respiratory acidosis with hypoxemia". Please review the grid.
- ☐ Screen 5 keys "Document chest tube drainage every shift" as indicated; after insertion drainage is usually checked/recorded hourly at first (bleeding). Consider rewording (e.g. "Measure and record drainage hourly").
- ☐ 1545 note: "Rales heard in the right lung fields" (the uninjured side) is unexplained; kept as "Crackles in the right lung fields". Also: chest tube "connected to 20 cm wall suction" now reads 20 cm H2O suction.
- ☐ SpO2 98% on room air at 1600, 15 minutes after the chest tube: oxygen removed quickly; confirm intended.
- ☐ ABG labels corrected (P02/PC02/SaP02/HCO3 → PaO2/PaCO2/SaO2/HCO3-); HCO3- 27 mEq/L → 27 mmol/L; gases stay in mm Hg. The ABG is charted at 1530 (the source gives no time). Temperatures shown in °C with °F.
- ☐ Screen 3 uses rationale scoring (1 point only if both blanks are correct), as in the source. Screen 4 options were reordered so a correct answer is not listed first.
- ☐ The bow-tie uses the chart at 1530 (as in the source); it overlaps screens 2-4, so it is a separate stand-alone.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS20 case_1790776800020

Screen 1 · Select all that apply

Which assessment findings require immediate follow-up? Select all that apply.

- ✓ Respiratory rate of 28/min
- ✓ SpO₂ of 89%
- ✓ Pain 5/10 with inspiration
- ✓ Absent breath sounds
- ✓ Cyanosis of the lips and fingertips

Screen 2 · Matrix (multiple per row)

For each client finding, click to specify if the finding is consistent with pulmonary contusion, cardiac tamponade, or tension pneumothorax. Each finding may be consistent with more than 1 condition. Each column must have at least 1 response option selected.

- Absent breath sounds → Tension Pneumothorax
- Asymmetrical chest movement → Tension Pneumothorax
- Respiratory acidosis with hypoxemia → Pulmonary Contusion, Tension Pneumothorax
- Hypotension → Pulmonary Contusion, Cardiac Tamponade, Tension Pneumothorax
- History of chest trauma → Pulmonary Contusion, Cardiac Tamponade, Tension Pneumothorax
- Muffled heart sounds → Cardiac Tamponade, Tension Pneumothorax
- Chest pain → Pulmonary Contusion, Cardiac Tamponade, Tension Pneumothorax
- Tracheal deviation → Tension Pneumothorax

Screen 3 · Dyad

Complete the following sentence by choosing from the lists of options.

- The client is most likely experiencing a [tension pneumothorax], as most evidenced by the [respiratory assessment].
- (Scored as a pair: 1 point only if both blanks are correct.)

Screen 4 · Select all that apply

Which of the following prescriptions should the nurse anticipate? Select all that apply.

- ✓ Assist with needle decompression
- ✓ Administer pain medication
- ✓ Obtain a STAT chest X-ray
- ✓ Set up a chest drainage system

Screen 5 · Matrix (one per row)

For each nursing action, click to specify if the action is indicated or not indicated.

- Administer morphine IV as needed → Indicated
- Position the drainage system at heart level → Not Indicated
- Ambulate in the hallway → Not Indicated
- Monitor fluctuations in the water seal chamber → Indicated
- Obtain a repeat chest X-ray in the morning → Indicated
- Document chest tube drainage every shift → Indicated
- Change the chest tube dressing every shift → Not Indicated

Screen 6 · Dyad

Complete the following sentence by choosing from the lists of options.

- The nurse determines the client's status is [improving]. The nurse should now [monitor breath sounds].

University of Maryland - CS20 Bowtie standalone_1790776801201

Question · Bow-tie

Complete the diagram by dragging from the choices below to specify what condition the client is most likely experiencing, 2 actions the nurse should take to address that condition, and 2 parameters the nurse should monitor to assess the client's progress.

- Condition: Tension pneumothorax
- Actions: Obtain a STAT chest X-ray; Call the rapid response team
- Parameters: Blood pressure; Breath sounds

University of Maryland - CS21: Thyroid Storm

Source: "Thyroid-Storm.doc" (medical-surgical) · Author: Suzana Jarquin, MSN, RN, CNEcl, Frederick Community College · In the bank, hidden (draft)

What to review

- ☐ Screen 5 answer key: the source .doc's highlighting could not be read, so the 2 orders to implement first were taken from the rationale (humidified oxygen and the 12-lead ECG). Please confirm.
- ☐ Oxygen is keyed as anticipated (screen 4) and as a first priority (screen 5) although SpO2 is 97% on room air; oxygen is not routinely indicated at that saturation. Consider lowering the SpO2 or re-keying.
- ☐ Treatment gaps: a beta-blocker (propranolol) is a cornerstone of thyroid storm treatment for the tachycardia and is not in the orders or options; propylthiouracil is often preferred to methimazole in thyroid storm (blocks T4-to-T3 conversion), and iodine and glucocorticoids are also usual. Consider adding propranolol to the orders/options.
- ☐ Screen 5 rationale said "Now medication orders should be implemented within 60-90 minutes"; removed (a STAT/now order is given promptly). The order "Admission" was written as "Admit to the hospital".
- ☐ Screen 6 asks about the laboratory report, which the source shows only in the bow-tie; the Laboratory Results (drawn at 0910) and the ECG report were added to screen 6's chart.
- ☐ Labs in SI with the author's ranges: WBC $13 \times 10^9/L$ (4.5-10.5); K 5.0 mmol/L; Na 142 mmol/L; TSH 0.27 mU/L (0.45-4.5); total T3 400 ng/dL → 6.1 nmol/L (80-200 ng/dL → 1.2-3.1 nmol/L); T4 270 nmol/L (57-148). In thyroid storm TSH is usually suppressed (< 0.01 mU/L); 0.27 is only mildly low. Consider changing it.
- ☐ Screen 1 ("What 4 client findings are most urgent?") is a select-4 question; anxiety is keyed not urgent. Screen 1 and screen 4 options were reordered so a correct answer is not listed first.
- ☐ The source uses they/them for the client in the notes and "female" in the intro; kept as in the source.
- ☐ Read the whole case in the studio preview (With answers) and check the rationales; then publish it (see page 1).

Answer key at a glance

University of Maryland - CS21 case_1790776800021

Screen 1 · Select N (4)

Which 4 client findings are most urgent?

- ✓ Dyspnea
- ✓ Tachycardia
- ✓ Palpitations
- ✓ Fever

Screen 2 · Matrix (one per row)

For each assessment finding, click to specify whether the finding is helpful or not helpful in determining if the client is experiencing a thyroid problem.

- Fever → Helpful
- Bilateral eye protrusion → Helpful
- Lung sounds → Not Helpful
- Anxiety level → Helpful
- Palpitations → Helpful
- Dyspnea → Helpful

Screen 3 · Drag and drop

Drag the most appropriate choice from the list of options to fill in the blank of the following sentence.

- The nurse recognizes that the condition the client is most likely experiencing is [thyroid storm].

Screen 4 · Select all that apply

Which of the following prescriptions should the nurse anticipate? Select all that apply.

- ✓ Humidified oxygen via nasal cannula at 2 L/min
- ✓ 12-lead ECG
- ✓ Oral acetaminophen
- ✓ Cooling blanket
- ✓ Oral methimazole

Screen 5 · Highlight

Click to highlight the 2 orders the nurse should implement first.

- ✓ Humidified oxygen at 2 L/min via nasal cannula
- ✓ 12-lead ECG

Screen 6 · Matrix (one per row)

For each finding, click to specify if the finding indicates that the client's status has improved or remains unchanged.

- Temperature → Improved
- Anxiety level → Unchanged
- Respiratory rate → Improved
- Heart rate → Improved
- Oxygenation → Unchanged
- Pain level → Unchanged

University of Maryland - CS21 Bowtie [standalone_1790776801211](#)

Question · Bow-tie

Complete the diagram by dragging from the choices below to specify what condition the client is most likely experiencing, 2 actions the nurse should take to address that condition, and 2 parameters the nurse should monitor to assess the client's progress.

- Condition: Thyroid storm
- Actions: Administer methimazole; Promote a calm, restful environment
- Parameters: Body temperature; Thyroid levels